Babu Middya

Machine Learning Engineer | AI Engineer

babumiddya804@gmail.com | +91 8327486160 | github.com/BM804 | linkedin.com/in/babu-middya-965400366

RESUME SUMMARY

Computer Science and Engineering graduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and AI-powered applications. Skilled in Python, data preprocessing, CNN-based model development, image classification, OCR, numerical computing, and 3D geometry processing. Experienced in developing ML projects involving handwritten character recognition and automated 3D model generation. Interested in building practical, scalable, and intelligent AI/ML solutions.

EDUCATION

Class X, *Sonamukhi B.J High School.* 2020
West Bengal Board of Secondary Education.
Percentage: 72%.

Class XII, *Sonamukhi B.J High School.* 2022
West Bengal Council of Higher Secondary Education.
Percentage: 90% .

B.Tech-Computer Science Engineering, *Techno Engineering College Banipur* 2022 – 2026
CGPA: 7.23/ 10

TECHNICAL SKILLS

Programming Languages:
C, Python, JavaScript

Data Science & Scientific Computing:
NumPy, Pandas, Matplotlib, Data Analysis, Data Visualization, Numerical Computing

Tools:
Git, GitHub, VS Code

Deep Learning
Convolutional Neural Networks (CNN), Image Classification, Handwritten Character Recognition, OCR

Machine Learning:
Machine Learning, Supervised Learning, Data Preprocessing, Dataset Preparation, Feature/Data Analysis, Model Training, Model Validation, Model Evaluation, Accuracy Analysis

Computer Vision
Image Processing, Image Resizing, Image Normalization, Noise Reduction, Image Classification, Visual Data Analysis

Databases:
MySQL, Database Design, Query Optimization

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations - Training Badge
July 03, 2025

AWS Academy Graduate - Cloud Foundations - Training Badge
June 28, 2025

PROJECTS

Bengali Handwritten character recognition
- Developed an ML-based OCR system for handwritten Bangla characters using CNN.
- Trained on a custom labeled dataset with preprocessing techniques including resizing, normalization, and noise reduction.
- Achieved **90% accuracy** on test/validation data.
- Designed the solution to be scalable and adaptable to multiple real-world applications, including document digitization, automated form processing, handwriting analysis, and educational systems.
- Demonstrates the practical application of CNN-based OCR for solving real-world document processing and language recognition problems across multiple domains.

MoldMaker – Automatic 3D Mold Generation
- Developed a Python-based AI/automation system that analyzes a 3D model and automatically generates a two-part mold design.
- Implemented 3D mesh analysis and geometric property extraction from the input model.
- Developed bounding-box calculation and parting-plane generation for automatic mold separation.
- Implemented computational geometry techniques for generating mold cavities.
- Generated separate top and bottom mold outputs for further 3D-printing workflows.
- Added 3D visualization and preview generation for inspecting the generated mold before export.
- Built a complete automated pipeline from **3D model input → geometry analysis → mold generation → visualization → output**.

AI-Based Automotive Welding Defect Detection
- Developed an object-detection system using YOLO11 to automatically identify and localize welding defects.
- Prepared and converted the LoHi-WELD dataset into YOLO format with train/validation/test splits.
- Trained and compared YOLO11n and YOLO11s models for four defect classes: **pore, deposit, discontinuity, and stain**.
- Evaluated model performance using **Precision, Recall, F1-score, mAP@50, and mAP@50–95**, and optimized the confidence threshold using validation results.
- Generated prediction visualizations and a **confusion matrix** for model analysis.